Girisankar Solaimalai

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SUMMARY

- MASc candidate and IMAE recipient with peer-reviewed contributions in aerosol transport, CFD, and fluid dynamics; experienced in model development, simulation, and experimental validation using OpenFOAM, ANSYS, MATLAB, and Python
- Cross-industry experience spanning R&D, process engineering, HVAC, mechanical design, manufacturing, and power generation
- Strong mechanical and mechatronics foundation spanning thermal-fluid systems, control systems, automation, and electromechanical design

EDUCATION

Master of Applied Science · Mechanical and Mechatronics Engineering *Sep 2025 – Aug 2027 (expected)*
University of Waterloo · Waterloo, ON
- Research in fluid dynamics and aerosol instrumentation technology · International Masters Award of Excellence (IMAE) · Coursework: CFD for Engineering Design, Advanced Applied Thermal Engineering, Advanced Fluid Mechanics, Convective Heat Transfer

Bachelor of Applied Science · Mechanical Engineering (Mechatronics Option) *Sep 2020 – May 2025*
University of Waterloo · Waterloo, ON
- GPA: 3.90/4.00 · Dean's Honours · President's Scholarship of Distinction (2020) · President's Research Award (2025)

EXPERIENCE

R&D Engineering Intern *Sep 2025 – Present*
Atmose · Edmonton, AB
- Developed a coupled multiphysics CFD model in OpenFOAM integrating charged aerosol transport and electrostatics to optimize Faraday cup electrometer geometry for minimal current leakage and maximum signal
- Investigated residence time distributions (RTD) in a radioactive neutralizer using CFD and experimental methods; redesigned internal flow geometry to reduce recirculation and improve neutralization performance
- First-authored two peer-reviewed manuscripts on Faraday cup electrometer induced currents and sampling line RTD; co-authored a third on radioactive neutralizer flow geometry optimization - all under review

Graduate Teaching Assistant *Sep 2025 – Present*
University of Waterloo · Waterloo, ON
- SYDE 381: Thermodynamics (Spring 2026) - ran weekly tutorials for 80+ Systems Design students; drafted assignment problems with full worked solutions and marked with written feedback
- ME 250: Thermodynamics 1 (Fall 2025) - led weekly tutorials and office hours for 100+ second year ME students; prepared assignments and marked using shared rubrics for consistent grading

Project Engineering Intern *Sep 2024 – Dec 2024*
Zeton Inc · Oakville, ON
- Performed transient thermal modelling in MATLAB to calculate heat-up times and size heat tracing systems (15–25 W/m) ensuring all piping and heat exchangers reach 600°C before introducing a proprietary molten salt to prevent solidification
- Developed and verified P&IDs for a carbon capture pilot plant (200+ instruments); co-authored the Cause and Effect matrix defining 80+ interlocks, alarms, and shutdown sequences

Mechanical Designer Intern *Jan 2024 – May 2024*
Smith and Andersen · Toronto, ON
- Designed mechanical systems including stormwater drainage in Revit for a $40 million luxury hotel; performed HVAC load analysis and equipment selection for a prospective 83-storey condominium
- Reviewed 200+ supplier shop drawings across 10+ projects, verifying fabricated equipment conformed to design specifications and engineering calculations

Manufacturing Body Weld Engineer Intern *May 2023 – Aug 2023*
Toyota Motor Manufacturing · Cambridge, ON
- Eliminated 63% of Side Member AGV faults by analysing PLC traffic counters to isolate overshoot as root cause, then designing, DFMA-optimizing, and machining an in-house mechanical engagement stopper
- Developed DELMIA weld robot programs for Kaizen cycle-time reduction and backup robot coverage to improve line reliability; built a Power BI maintenance dashboard for real-time fault visibility

Project Manager Intern *Sep 2022 – Dec 2022*
Hubhead Corp. · Toronto, ON
- Restructured critical asset hierarchies for a large oilfield operator in Hubhead's ERP-integrated platform, extracting and verifying assets from P&ID drawings to enable centralized equipment visibility, maintenance scheduling, and BOM management

Mechanical Engineer Intern *Jan 2022 – May 2022*
Atura Power · Greater Napanee, ON
- Coordinated 20+ plant outage activities including cooling tower plume abatement and turbine blade replacement at Atura Power's 900 MW Napanee Generating Station; authored daily outage progress reports for executive stakeholders

Network Engineer Assistant *May 2021 – Aug 2021*
OmniCloud · Dubai, UAE
- Provided network troubleshooting and device initialization support; maintained technical documentation for connectivity infrastructure

TECHNICAL SKILLS

Simulation: MATLAB, Simulink, ANSYS (Fluent, CFX), OpenFOAM
Automation: PLC programming, DELMIA, C/C++, Python, ROS 2, FPGA
Design & CAD: SolidWorks, Siemens NX, Onshape, CATIA, Revit
Methods: DFMA, DFMEA, GD&T (ASME Y14.5), Rapid Prototyping